INCH-POUND

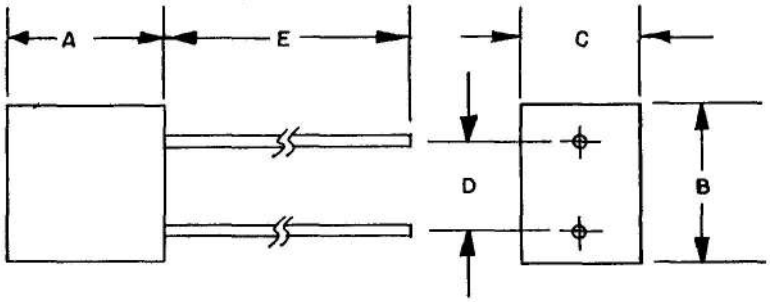
MS21422D
W/AMENDMENT 1
15 July 2022
SUPERSEDING
MS21422D
31 August 2007

MILITARY SPECIFICATION SHEET

COILS, RADIO FREQUENCY, MOLDED, FIXED, SUB-MINIATURE, TYPES LT4K496 TO LT4K520 INCL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification and MIL-PRF-15305.



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	Dimensions in inches with metric equivalents (mm) in parentheses
	Minimum	Maximum
A	.200 (5.08)	.230 (5.84)
B	.190 (4.83)	.210 (5.33)
C	.090 (2.29)	.110 (2.79)
D	.090 (2.29)	.110 (2.79)
E	1.00 (25.40)	----

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Dimensions and configuration.

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FSC 5950

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REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1.

Style: LT4

Grade: 1
Class: B

Weight: .5 gram, maximum.

Operating temperature range: -55° to +125°C.

Ambient temperature: + 90°C maximum.

Temperature rise: 35°C.

Power dissipation: 0.2 watt, maximum.

Lead wire: Lead wire shall be type LW (0.51) C-xx-N, (where xx is Final Finish 32,,34, 41, 43, 45, 51, or 52) in accordance with MIL-STD-1276.

Terminal pull: 5 pounds minimum.

Altitude: 70,000 feet.

Shock, specified pulse: MIL-STD-202-213, test condition I, is applicable.

Vibration (high frequency): MIL-STD-202-204, test condition D.

Dielectric withstanding voltage:

At sea level: MIL-STD-202-301, test voltage 700 V rms minimum.

At reduced barometric pressure: MIL-STD-202-105, test condition C, test voltage 200 V rms minimum.

Electrical characteristics: See tables I and II.

Inductance: See table I.

Q values: See table I.

Self-resonant frequency (SRF): See table I.

DC resistance (DCR): See table I.

Part or Identifying Number (PIN): MS21422 - (dash number from table I).

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TABLE 1. Electrical characteristics (initial).

1/ Dash Number MS21422	Type Designa-tion	Inductance (μH)	Q (min)	Test frequency (MHz)	SRF min (MHz)	DC resistance (max) (ohms)	Max rated DC current (mA)
-01	LT4K496	.10 ±10%	55	25	450	.04	2200
-02	LT4K497	.12 ±10%	60	25	400	.05	2000
-03	LT4K498	.15 ±10%	60	25	350	.06	1800
-04	LT4K499	.18 ±10%	60	25	320	.07	1600
-05	LT4K500	.22 ±10%	65	25	300	.08	1500
-06	LT4K501	.27 ±10%	65	25	280	.10	1400
-07	LT4K502	.33 ±10%	65	25	260	.11	1300
-08	LT4K503	.39 ±10%	65	25	240	.14	1200
-09	LT4K504	.47 ±10%	65	25	220	.17	1100
-10	LT4K505	.56 ±10%	70	25	200	.22	1000
-11	LT4K506	.68 ±10%	70	25	180	.27	900
-12	LT4K507	.82 ±10%	70	25	160	.30	800
-13	LT4K508	1.0 ±5%	70	25	150	.35	750
-14	LT4K509	1.2 ±5%	60	7.9	130	.40	700
-15	LT4K510	1.5 ±5%	60	7.9	120	.50	630
-16	LT4K511	1.8 ±5%	60	7.9	110	.70	530
-17	LT4K512	2.2 ±5%	60	7.9	100	.90	470
-18	LT4K513	2.7 ±5%	60	7.9	90	1.1	420
-19	LT4K514	3.3 ±5%	60	7.9	70	1.3	390
-20	LT4K515	3.9 ±5%	60	7.9	60	1.5	360
-21	LT4K516	4.7 ±5%	60	7.9	50	1.8	330
-22	LT4K517	5.6 ±5%	60	7.9	45	2.0	310
-23	LT4K518	6.8 ±5%	60	7.9	40	2.2	300
-24	LT4K519	8.2 ±5%	60	7.9	37	2.4	290
-25	LT4K520	10.0 ±5%	60	7.9	35	2.6	280

1/ The dash number added to the MS military standard number constitutes the MS part number, for example MS21422-01

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TABLE II. Electrical characteristics (final). 1/

Inspection group	Allowable variation from Initial measurement	Allowable variation from Initial measurement	Allowable percent from specified minimum value in electrical characteristics (initial) table	Allowable percent from specified minimum value in electrical characteristics (initial) table
	Inductance (percent)	DC resistance	Self-resonant frequency	Q
Qualification inspection				
Group II	±2	---	---	-10
Group III	±5	±(3% +.001 ohm)	-8	-10
Group IV	±5	±(2% +.001 ohm)	-10	-10
Conformance inspection group C				
Subgroup I	±2	---	---	-10
Subgroup II	±5	±(2% +.001 ohm)	-10	-10
Subgroup III	±5	±(3% +.001 ohm)	-8	-10

Application notes:

1. The overload current shall be 1.4 times the rated value specified in table I.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-15305, this document references the following:

MIL-STD-202-105 MIL-STD-202-204 MIL-STD-202-213 MIL-STD-202-301 MIL-STD-1276

Custodians:
Army – CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Air Force – 19

Preparing activity:
DLA – CC

(Project 5950-2022-051)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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